

A SEVEN-VERTEX BIPARTITE GRAPH WHOSE POSITIVE MATCHING DECOMPOSITION NUMBER LIES BELOW ITS SLOPE INVARIANT

ABSTRACT. Farrokhi D. G., Gharakhloo and Yazdan Pour introduced the positive matching decomposition number $\text{pmd}(\Gamma)$ of a graph and, for a bipartite graph, an invariant $K(\Gamma)$ obtained by minimising the output of a greedy slope-based algorithm over all labellings of the two sides; they proved $\Delta(\Gamma) \leq \text{pmd}(\Gamma) \leq K(\Gamma)$ and asked whether the second inequality is ever strict. We answer this affirmatively with the theta graph $\Gamma = \Theta(2, 2, 4)$ on seven vertices: $\text{pmd}(\Gamma) = 3$ by a decomposition exhibited below and checkable by hand, while $K(\Gamma) = 4$ by running the authors' own algorithm on all $3!4! = 144$ labellings (all 288, if the two sides may also be exchanged).

1. THE QUESTION

Let Γ be a finite simple graph. Following Farrokhi D. G., Gharakhloo and Yazdan Pour [1], a matching M of Γ is *positive* when the corresponding sum of variables generates a prime ideal after inversion; we never use that definition directly, only the combinatorial characterisation the authors prove for it [1, criterion (iii)], which we therefore take as the working definition here:

- a matching M of Γ is *positive* if and only if for every non-empty
- (1) $N \subseteq M$ the induced subgraph $\Gamma[V(N)]$ on the endpoints of N has a vertex of degree 1 whose incident edge lies in N .

The *positive matching decomposition number* $\text{pmd}(\Gamma)$ is the least p for which the edge set admits an *ordered* partition $E(\Gamma) = M_1 \uplus \cdots \uplus M_p$ such that each M_i is a positive matching of $\Gamma - (M_1 \cup \cdots \cup M_{i-1})$. Since a matching contains at most one edge at each vertex and the parts cover $E(\Gamma)$,

- (2)
$$\Delta(\Gamma) \leq \text{pmd}(\Gamma).$$

For a bipartite graph the authors introduce an upper bound through a greedy algorithm driven by a numerical *slope*. Fix a bipartition $V(\Gamma) = X \uplus Y$ and a labelling τ , that is, an ordering $X = \{x_1, \dots, x_m\}$ and $Y = \{y_1, \dots, y_n\}$, and put $s(x_i y_j) = j - i$. Their Algorithm 1 produces parts $M_1, M_2, \dots$ from the edge set; we quote it from [1], with the pseudocode macros set as words and indentation.

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REQUIRE: A labeled bipartite graph Γ with bipartition X, Y
 ENSURE: A positive matching decomposition of Γ
 $k \leftarrow 0$
while $E(\Gamma) \neq \emptyset$
 $k \leftarrow k + 1$
 $X' \leftarrow X \setminus \{\text{isolated vertices of } X\}$
 $Y' \leftarrow \emptyset$
 $M_k \leftarrow \emptyset$
 while $X' \neq \emptyset$
 $s \leftarrow \min(\text{Slopes}(\Gamma[X' \cup Y]))$
 $M \leftarrow \{xy \in E(\Gamma[X' \cup Y]) : \text{slope}(xy) = s\}$
 $M_k \leftarrow M_k \cup \{xy \in M : y \notin Y'\}$
 $X' \leftarrow X' \setminus \{x : xy \in M\}$
 $Y' \leftarrow Y' \cup \{y : xy \in M\}$
 end while
 $E(\Gamma) \leftarrow E(\Gamma) \setminus M_k$
end while
return $M_1, \dots, M_k$

Two edges of equal slope s have distinct X -endpoints and, since $j = s + i$, distinct Y -endpoints, so the class M selected at a pass of the inner loop is itself a matching and the order in which its edges are examined is immaterial: each part, and hence the number $K(\tau)$ of parts, is well defined. Each part is a matching, since an x leaves X' as soon as it contributes and a Y -endpoint is used at most once. Finally

$$(3) \quad K(\Gamma) = \min_{\tau} K(\tau),$$

the minimum over all $m!n!$ labellings, and [1] proves $\text{pmd}(\Gamma) \leq K(\Gamma)$; with (2) this gives the chain $\Delta \leq \text{pmd} \leq K$ that precedes the question we answer. The source denotes this invariant by a fraktur $\mathfrak{K}$; we use an upright K . The question is [1, Question 1]:

Question 1. Is there any bipartite graph Γ satisfying $\text{pmd}(\Gamma) < K(\Gamma)$?

Since K is defined through the slope $j - i$, which is not symmetric in the two sides, and since Algorithm 1 peels from X , the definition leaves open whether a witness may also exchange X and Y . Every value below is given for both readings: K minimised over the $m!n!$ labellings of one orientation, and over all $2m!n!$ labellings when the sides may be exchanged. Nothing in this paper depends on the choice.

2. THE WITNESS

Let $X = \{x_1, x_2, x_3\}$, $Y = \{y_1, y_2, y_3, y_4\}$ and

$$(4) \quad E(\Gamma) = \{x_1y_1, x_1y_2, x_2y_1, x_2y_3, x_2y_4, x_3y_2, x_3y_3, x_3y_4\},$$

that is, the rows of the bipartite adjacency matrix are the bitmasks 3, 13, 14 over $y_1, \dots, y_4$. So Γ has 7 vertices and 8 edges, degrees 2, 3, 3 on X and

2, 2, 2, 2 on Y , and $\Delta(\Gamma) = 3$. The two vertices of degree 3 are x_2 and x_3 , and deleting them leaves three paths with 1, 1 and 3 vertices, so

$$\Gamma = \Theta(2, 2, 4),$$

the graph obtained from two vertices by joining them with three internally disjoint paths of lengths 2, 2 and 4. Under the labelling $0, 1, 2 = x_1, x_2, x_3$ and $3, 4, 5, 6 = y_1, y_2, y_3, y_4$ its graph6 string is FEhb?.

Theorem 1. *For $\Gamma = \Theta(2, 2, 4)$ as in (4),*

$$\text{pmd}(\Gamma) = 3 < 4 = K(\Gamma),$$

the value of K being 4 both over the 144 labellings of the orientation printed above and over all 288 labellings when the two sides may be exchanged. In particular Question 1 of [1] has answer yes.

The two halves are of different character and we treat them separately: the value of pmd is settled by an exhibited decomposition and (2), the value of K by a finite exhaustion of the authors' own algorithm.

$$\text{pmd}(\Gamma) = 3.$$

Lemma 2. *Let M be a matching of a graph Γ such that $\Gamma[V(M)]$ is acyclic. Then M is positive.*

Proof. Let $\emptyset \neq N \subseteq M$ and $F = \Gamma[V(N)]$. As $V(N) \subseteq V(M)$ and an induced subgraph of an induced subgraph is induced, F is a subgraph of $\Gamma[V(M)]$ and hence a forest. Every edge of N has both endpoints in $V(N)$, so $N \subseteq E(F)$ and F has an edge; a forest with an edge has a vertex ℓ of degree 1. Now $\ell \in V(N)$ and N is a matching, so ℓ is an endpoint of exactly one edge $f \in N$, and $f \in E(F)$. Since ℓ has degree 1 in F , the unique edge of F at ℓ is f . Thus F has a degree-1 vertex whose incident edge lies in N , which is (1). $\square$

Corollary 3. *If Γ is a forest then every matching of Γ is positive and $\text{pmd}(\Gamma) = \Delta(\Gamma)$.*

Proof. Induced subgraphs of a forest are forests, so Lemma 2 applies to every matching. A forest is Δ -edge-colourable, and each colour class is a matching, hence a positive one; so $\text{pmd} \leq \Delta$, and (2) gives equality. $\square$

Proof that $\text{pmd}(\Gamma) = 3$. Put

$$M_1 = \{x_2y_1, x_3y_3\},$$

$$M_2 = \{x_1y_1, x_2y_4, x_3y_2\},$$

$$M_3 = \{x_1y_2, x_2y_3, x_3y_4\},$$

an ordered partition of the eight edges (4) into three matchings. The subgraph induced by $V(M_1) = \{x_2, y_1, x_3, y_3\}$ is the path $y_1x_2y_3x_3$, which is acyclic, so M_1 is positive by Lemma 2. Deleting M_1 leaves exactly the path

$$\Gamma - M_1 = y_1 x_1 y_2 x_3 y_4 x_2 y_3$$

on all seven vertices, a tree with $\Delta = 2$; by Corollary 3 its edge set splits into two positive matchings, namely its two alternating classes, which are M_2 and M_3 . Hence $\text{pmd}(\Gamma) \leq 3$. Conversely $\text{pmd}(\Gamma) \geq \Delta(\Gamma) = 3$ by (2), so $\text{pmd}(\Gamma) = 3$. $\square$

$K(\Gamma) = 4$. Algorithm 1 is deterministic once a labelling is fixed, and there are only $3!4! = 144$ labellings per orientation, so the value (3) is a finite computation on a graph with 8 edges. Carried out over all of them, the number of parts takes the values $\{4, 5\}$ on the orientation printed in (4) and the constant value 4 on the orientation with X and Y exchanged. Hence

$$K(\Gamma) = 4$$

in both readings, and with $\text{pmd}(\Gamma) = 3$ this proves Theorem 1. For instance the printed labelling itself gives the four parts $\{x_2y_1, x_3y_2\}$, $\{x_1y_1, x_3y_3\}$, $\{x_1y_2, x_2y_3, x_3y_4\}$, $\{x_2y_4\}$. The exhaustion is carried out by `verify.py` in exact integer arithmetic.

3. VERIFICATION

The value $\text{pmd}(\Gamma) = 3$ can be checked by hand from what is printed above: the eight edges (4), the three parts M_1, M_2, M_3 , Lemma 2 and the bound (2). The value $K(\Gamma) = 4$ cannot. Its lower bound is the exhaustion of the 144, respectively 288, labellings in (3), and we have no argument for it that a reader can carry out without a machine.

The accompanying program `verify.py` (Python 3.9 or later, standard library only, no external data file) reads the objects printed here as literals and re-derives every quantity this paper states: the decoding of the graph6 string `FEhb?` and its agreement with (4); the identification of Γ as $\Theta(2, 2, 4)$; the agreement of two structurally independent implementations of (1); the positivity of M_1, M_2, M_3 in their successive residuals and the value $\text{pmd}(\Gamma) = 3$ by exhaustive search; and $K(\Gamma) = 4$ over 144 and over 288 labellings. It prints one line per check, closes with `VERDICT: ALL 80 CHECKS PASS` and exits 0 only if every check passed.

The recorded run establishes more than this paper claims, and its labelling was not brought back into line with the present text. Steps 1–6 are the checks that bear on Theorem 1. Steps 7 and 8 — a census of the bipartite graphs of order at most 7, the minimality and uniqueness of Γ among them, and a second, cubic witness M_5 on ten vertices — concern statements this paper does not make, and their labels, and the closing `NOTE SCOPE` paragraph, speak of values “the paper prints” that are printed in no version of this note. Those labels are inherited from a longer manuscript and should be read as claims about the program’s own computation, not about anything asserted here. Nothing in Sections 1–2 depends on them, and this note asserts none of them.

Two things are taken from [1] rather than checked here. The characterisation (1) is a theorem of the source, used as the definition of “positive”

throughout; nothing here returns to the algebraic definition. And Algorithm 1 has to be read: the program implements it both as the source's pseudocode, statement for statement in the source's own variables, and in the streamlined form used for the exhaustions, checks the two against each other, and finds that both reproduce, part for part, the parts the source prints for each of its two worked examples; but that the statements displayed in Section 1 are the ones printed at lines 549–574 of the e-print source of [1] is a comparison left to the reader.

REFERENCES

- [1] M. Farrokhi D. G., Sh. Gharakhloo, and A. A. Yazdan Pour, *Positive matching decompositions of graphs*, Discrete Appl. Math. **320** (2022), 311–323. doi:10.1016/j.dam.2022.05.012; arXiv:2110.12168. Statement numbers and line references are quoted from the e-print v2, the only full text we could access.